

암 dark matter 의 분자 sub-stratification

An 8-gene RAI panel resolves a fusion-driven, epigenetically silenced subtype within BRAF/RAS-negative papillary thyroid carcinoma

~23%의 BRAF/RAS-음성 PTC dark matter (Xing 2014 NEJM 명명) 안에서 **76.8% tyrosine-kinase fusion enrichment + fusion-independent promoter hypermethylation (TPO Cohen's $d = 2.30$)** 을 동시에 설명하는 8-gene 패널 + DM1/DM2 cluster. TCGA-THCA $n=504$ + MSK-IMPACT $n=117$ + Korean K2/Lee $n=865$ + 외부 single-cell (Pu 2021 · Lu 2023) 5-pillar evidence chain. **2026-05-04 기준 status:** Title Cand 1 ★ + Abstract v2 + Outline v3 확정 · Cross-contamination audit PASS post-mechanical-edit · W1–W6 marathon 5/7 진입 예정.

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§ 1

Executive summary ★

5 paper-defining 신호 + 현재 진척률.

76.8%

DM1 FUSION-
POSITIVE ★

RET 33 · NTRK 10 ·
ALK 4 · BRAF 5 · OR
7.41 vs DM2 ·
cBioPortal SV n=542

2.53

POOLED OS HR ★

[1.31, 4.89] $I^2=0\%$ ·
TCGA + MSK random-
effects DerSimonian-
Laird

2.30

TPO PROMOTER D ★

DM1 vs DM2
hypermethylation ·
mean 8-gene β 0.385
vs 0.253 · fusion-
independent

81.8%

RET+ CAPTURED BY
DM1

27/33 TCGA RET-
fusion → DM1 reflex
algorithm · 48
selpercatinib-
eligible/1000 PTC

0.92

PAN-GENOME ARI

top-5000 ≈ TIERA67
0.90 · 8-gene alone
0.49 · panel-driven
artifact 차단

-0.04

BRAF MRNA D
(V600E VS WT)

driver-mutation
neutrality · panel
mechanism not
confounded

874

KOREAN COHORTS

K2 235 + Lee 630 +
GSE286332-PTC 9
(advisor decision
pending option a/b)

PASS

CROSS-
CONTAMINATION
AUDIT

Title Cand 1 ★ +
Abstract body PASS ·
Outline 4 HARD post-
mechanical-edit clean

PAPER-DEFINING 한 줄

DM1 = fusion-driven, epigenetically silenced subtype within BRAF/RAS-negative dark matter — RET reflex algorithm + HMA + RAI re-induction rationale. 5 pillars 가 동일 DM1 sub-stratum 을 5가지 직교 layer (orthogonal axis · clinical aggressiveness · driver neutrality · cross-cohort transfer · two-mechanism integration) 로 동시 충족. Pillar 5 (fusion + epigenetic) 가 paper-defining novelty.

★ TITLE CAND 1 – MECHANISM-FORWARD (RECOMMENDED, 99 CHARS)

*"Fusion-driven, epigenetically silenced dark matter in
BRAF/RAS-negative papillary thyroid carcinoma"*

왜 본인이 이 paper 를 쓸 수 있고, 왜 지금이어야 하는가.

2.1 Clinical motivation

분화 갑상선암의 5년 생존율은 98% 이상이지만, ATA 2015 intermediate-risk 환자의 약 20% 에서 구조적 재발이 발생합니다 (Haugen et al. 2016). BRAF/RAS-음성 dark matter (~23%) 는 이 재발의 핵심 미해결 영역으로, 현재 RAI 치료 결정은 임상-병리 변수와 BRAF V600E 단일 marker 에만 의존하고 있습니다. ATA 2025 (Ringel et al. 2025) 에도 fusion driver 와 epigenetic silencing 은 risk tier 에 반영되지 않은 상태 — 본 paper 가 이 공백을 정확히 충족합니다.

2.2 본인 자산 정렬

본 분석은 본인의 기존 도구 chain 을 정확히 활용합니다 — (i) **TIERA67** 67-gene candidate pool 정의 (driver gene 미포함, biology-curated) · (ii) **DM1/DM2** classifier (Cancer paper Phase 0) · (iii) cBioPortal API 기반 structural variant + HM450 methylation 통합 처리. 두 분 (주영석 / 최정균) 도 갖고 있지 않은 본인 고유 자산.

2.3 Convergence framing — Yoo 2016 + Landa 2016

8-gene panel 은 **Yoo SK et al. 2016 PLOS Genet** 의 RAI uptake biology 16-gene TDS-core 의 subset (panel access 시점 Landa 2016 ATC 데이터 미접근). **Landa I et al. 2016 JCI 126(3):1052-1066** 의 advanced thyroid cancer (84 PDTC + 33 ATC) genomic landscape 에서 분화 transcript (TG · TSHR · TPO · PAX8 · SLC26A4 · DIO1 · DUOX2) 가 ATC 끝단에서 profoundly suppressed. **5/8 gene overlap = independent design 이 동일 biology 에 도달한 convergent validation** (reverse-causality 차단 3-layer: Methods Yoo 2016 first cite · Discussion 3.1 convergence framing · Cover Letter explicit statement).

§ 3

5-Pillar evidence chain

5 직교 layer 가 동일 DM1 sub-stratum 을 동시 충족 — *orthogonal axis · clinical aggressiveness · driver neutrality · cross-cohort transfer · two-mechanism integration.*

I

DM1/DM2 discovery

8-gene panel +
leiden K=2 ·
TIERA67-derived
· ΔAUC 8 vs 16 =
0.013 NS

STRONG

II

Clinical HR meta

TCGA HR 2.30 +
MSK 2.67 ·
pooled 2.53
[1.31, 4.89] $I^2=0\%$

STRONG

III

Driver neutrality

BRAF mRNA
 $d=-0.04$ V600E
vs WT · pan-
genome ARI 0.92

STRONG

IV

Cross-cohort sc

Pu 2021 6/6
 $r>0.79$ · Lu 2023
 $r=0.97$ (autocorr
disclosure)

STRONG

V ★

Fusion + epigenetic

76.8% fusion+ ·
TPO $d=2.30$
promoter
hypermethylation
fusion-
independent

STRONG ★

Pillar I — 8-gene panel + DM1/DM2 cluster discovery

STRONG · 5/4 본인 VOICE 영역 §2.1 · FIG 1 (4 PANELS) · V4 §5.1, P1 + P4

TIERA67 67-gene candidate pool — 7 biology category curation, driver gene 미포함 (panel-driven artifact 차단). 8-gene = TDS-core subset (Yoo 2016 PLOS Genet RAI uptake 16-gene 의 8 / 16).
leiden K=2 unsupervised cluster → DM1 (MAPK-active, dedifferentiated) + DM2 (thyroid-lineage-preserved). ΔAUC (8 vs 16) = 0.013 NS — **parsimony principle 충족**.

Pillar II — Clinical aggressiveness + Meta-analysis

STRONG · FIG 2 + FIG 6 FOREST · V4 §5.4, R2-N1

Xing 2014 dark matter 의 73% sub-stratification rescue (180 BRAF-/TERT- → 131 sub-stratified).
TCGA HR 2.30 [0.77, 6.88] + MSK HR 2.67 [1.17, 6.10] → DerSimonian-Laird random-effects
pooled **HR 2.53 [1.31, 4.89]**, $I^2=0\%$, **Cochran Q p=0.79**. Heterogeneity 없는 meta-analysis 가
reviewer 의 "single-cohort artifact" 우려 사전 차단.

Pillar III — Driver mRNA neutrality + pan-genome reproducibility

STRONG · FIG 1 PANELS C+D · GRAND P1, P4

BRAF transcript Cohen's $d = -0.04$ (V600E carriers vs wild-type) — **panel mechanism 0 | driver mutation status 와 independent**. Pan-genome top-5000 features ARI = 0.92 $\approx$ TIERA67 0.90; 8-gene alone ARI = 0.49; Driver_anchor only ARI = -0.007 . **"Candidate-pool restriction does not artificially impose this structure"** — reviewer 의 "panel trivial?" Q 사전 차단.

Pillar IV — Cross-cohort + Single-cell external validation

STRONG · FIG 3 + FIG 5 · V4 §5.5, R4-3, D8-B

Single-cell external — **Pu et al. 2021 Nat Commun 12:6058 (GSE184362, 6 PTC patients)** $r = 0.798\text{--}0.886$ thyrocyte-intrinsic DM1 signal · **Lu 2023** $r = 0.97$ (panel autocorrelation $\Delta d = 0.30$ honest disclosure). FFPE robustness KS $p = 0.44$. Korean K2 235 + Lee 630 (advisor option a/b 결정 후 GSE286332-PTC 9 포함 여부) — bulk RNA-seq cross-population transfer.

Pillar V ★ — Fusion-driven + epigenetically silenced dark matter

PAPER-DEFINING · FIG 7 + FIG 8 · V4 §5.0 R3-F4 + R5-2

5.1 — Driver-fusion enrichment

DM1 76.8% fusion-positive (63/82): RET 33 · NTRK 10 · ALK 4 · BRAF 5. DM1 vs DM2 OR 7.41 (sensitivity 7.18–9.07). MSK SV cross-validation (RET 5 · ALK 3 · PAX8-PPARG 3). Missingness MAR (chi² p=0.56). cBioPortal SV n=542 (542/557 SV-tested).

5.2 — Promoter hypermethylation (fusion-independent)

Cohen's d table — **TPO 2.30 · DIO1 1.24 · TSHR 1.20 · PAX8 0.97 · TG 0.86 · FOXE1 0.84 · NKX2-1 0.63** · SLC5A5 NS d=0.22. Mean 8-gene β **0.385 (DM1) vs 0.253 (DM2)**. **Within-DM1 fusion+ vs fusion- d = -0.36 NS — methylation 은 fusion driver 와 독립적 layer**. 3-layer model (L1 genetic · L2 heterogeneity · L3 epigenetic).

5.3 — Translational implication

RET reflex algorithm — DM1 captures 81.8% TCGA RET-fusion (27/33). Population estimate **48 selpercatinib-eligible / 1000 PTC** (Wirth 2020 NEJM LIBRETTO-001 ORR 79%). HMA + RAI re-induction rationale = TPO/DIO1/TSHR promoter hypermethylation epigenetic-targeted (decitabine retrospective ATC trials NCT00085293, NCT01065090). SLC5A5/NIS exception → combination strategies. **"Evaluation, not trial"** dial-back tone.

§ 4

Title · Abstract · Outline status

Source file v3 확정 + audit PASS post-mechanical-edit.

Table 1 — Source file status (5/4 기준)

FILE	VERSION	STATUS	WORDS / LINES	본인 VOICE 영역
00_title_candidates.md	v2 (★ Cand 1)	LOCK pending	75 lines	Cand 1 → 5/6 lock

FILE	VERSION	STATUS	WORDS / LINES	본인 VOICE 영역
01_abstract.md	v2 (153w)	PASS	153 / 73	Conclusions (b) confirmed
02_outline.md	v3	PASS post-edit	328 lines	§3.1, §3.4 본인 키보드
03_introduction.md	v1 draft 715w	DRAFT	715 / 87	§1.1 Hook 첫 줄, §1.4 Aim
04_results.md	—	TBD	—	5/7 marathon W1 진입
06_discussion.md	—	TBD	—	§3.1 첫 paragraph 본인 키보드
07_star_methods.md	—	TBD	—	scaffolding Claude 보조
08_cover_letter.md	—	TBD	—	본인 voice 전체

§ 5

Cross-contamination audit summary

5/4 isolated session 산출물 — 자세한 결과는 [p1_cross_contamination_audit.html](#) 참조.

PASS TITLE BODY <i>Cand 1 + Cand 2 · 0 forbidden term</i>	PASS ABSTRACT BODY <i>Lines 11–19 · Conclusions (b) clean</i>	4 OUTLINE HARD <i>§2.4b · §2.5 · §3.3 · §3.4(iii)</i>	7 A1–A7 ACTIONS <i>~22 min mechanical edit post-approval</i>
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3 source file × 15 forbidden term grep 결과 — Title Cand 1 ★ + Abstract body 는 즉시 PASS. Outline §2.4b · §2.5 · §3.3 · §3.4(iii) 4 cell 에 prose-propagation risk HARD violation. **Pillar 5 boundary 의 의도 (Paper 1 = "older + immune-hot DM1 sub-B is mechanistically distinct" 까지) 는 take-home line 112 에서 깔끔하 나, §2.4b cell line 104 자체에 Paper 2 territory 통계 5개가 inline.** A1–A4 mechanical edit 후 §2.4b · §2.5 · §3.3 · §3.4 prose drafting 가능. 자세한 violation quote + recommended sanitization 은 별도 audit briefing 참조.

Open gaps + risks

3 open gaps · advisor 답장 전 본인 결정 필요.

1

GSE286332 PTC(9) inclusion in n=874

MED 01_ABSTRACT.MD:15,49 · 02_OUTLINE.MD:106 · A2 + A5 COHORT DECISION

Option (a) keep n=874 + STAR Methods caveat (statistical power 보존, boundary risk) vs **Option (b) recompute n=865 K2+Lee only** (cancer-only scope 강화, 1% power loss). 본인 권장 (b). Figure caption n 표기 일관 변경 ~10 분.

2

K2 mini-index calibration mismatch (D7-P3)

MED §3.4(III) LIMITATIONS · A4 REPHRASING

K2 8-gene mini-index TPM ~10–100× inflation (memory `v17_korean_k2_calibration.md`). 해결 — within-sample-centered profile (TCGA AUC 0.968). §3.4(iii) Limitations 정직 disclosure 필수. K2 alternate evidence (94.6% DM2 + sub-stratification fraction) 로 cover.

3

Bundang prospective cohort 0%

MED §3.4(II) LIMITATIONS · 분당 SNUH 응답 대기

현재 prospective biopsy cohort 0건. East-Asian 4,300+ retrospective tumor 로 cover (Korean DM 37.8% vs TCGA 28.4%). venue ladder "stretch" 표현 — Cell Rep Med ★ default, JCI Insight + Nat Commun reach, npj Precision Oncology fallback.

Marathon timeline · W1–W6

5/4 advisor 답장 → 5/5 prep day → 5/6 Title lock + Hook → 5/7–6/13 marathon W1–W6.

● 2026-04-30 분석 freeze · Marathon 모드 진입

memory v17_marathon_mode_post_pillar1.md · "Pillar 1 STRONG = 분석 끝. D9+ sprint 없음. 5/4-6/13 6주 manuscript writing marathon."

● 2026-05-04 AM Cross-contamination audit 완료 · A1-A7 식별

3 source file × 15 forbidden term grep · 4 HARD + 3 SOFT/META · advisor briefing webpage 생성.

● 2026-05-04 PM Advisor 답장 + A1-A4 confirm + GSE286332 option a/b 결정

한 줄 답: "A1-A4 confirm / A2 option B / A6-A7 sanitize 진행" → mechanical edit 권한 획득.

○ 2026-05-04 PM Outline · Abstract · Title-candidates mechanical edit (~22분)

A1-A7 적용 · Claude Code 가 본인 voice 보호 영역 미접근.

○ 2026-05-05 본인 prep day — Landa 1.5h + Yoo 30min + ATA 1-1.5h

Discussion 3.1 (Landa 2016) · Yoo 2016 PLOS Genet methods · ATA 2015+2025 fusion/epigenetic 반영 시나리오 1/2/3.

○ 2026-05-06 Title Cand 1 ★ lock + Hook 첫 단락 본인 키보드

"Fusion-driven, epigenetically silenced dark matter ..." 확정 · §1.1 Hook (voice-protected) 본인 작성.

○ 2026-05-07 → 2026-05-17 W1-W2 · Methods + Suppl freeze

Cell Press STAR Methods 5 sub-section · Suppl Tables S1-S15 · cohort assembly · panel selection · DM cluster · sc validation · survival meta · cBioPortal API · Korean processing · software.

○ 2026-05-18 → 2026-05-31 W3-W4 · Results + Discussion draft

§2.1-§2.5 Results 5 sub-result × 500-800w · §3.1-§3.4 Discussion 1,000-1,500w · §3.1 첫 paragraph + §3.4 Limitations 본인 voice.

○ 2026-06-01 → 2026-06-13 W5-W6 · Cover letter + submission

Cell Reports Medicine submission portal · cover letter 본인 voice · figure final SVG · response to reviewers prep.

Discussion questions for advisor

미팅 안건 — 본인 답 prepared.

Q1 "GSE286332 PTC arm 9명을 Paper 1 cohort 로 포함할 것인가? (option a vs b)"

본인 권장 option (b): n=865 (K2+Lee only) — cancer-only scope 강화, Paper 2 backbone 무관계, 1% power loss 미미. Figure caption n 표기 일관 변경 ~10 분.

Q2 "Title Cand 1 (mechanism-forward) vs Cand 2 (translational-forward) — venue 별 어떤 게 fit?"

본인 추천 Cand 1. Cell Reports Medicine 의 mechanism-forward 선호 + Nat Commun reach 정당화. Cand 2 는 npj Precision Oncology fallback 시 고려.

Q3 "Pillar 5 boundary — §2.4b 에서 Paper 2 hook 어디/까지 명시?"

Take-home phrase "warranting separate inquiry" 까지만 (line 112). Cell 안의 Paper 2 statistic 5개 (mediation · HLA n=874 · BCR · NBNR · 침윤 통계) 는 A1 mechanical edit 으로 drop.

Q4 "Discussion 3.2 의 ATA 2025 fusion/epigenetic 반영 여부 시나리오 1/2/3 중 어느 것?"

5/5 본인 ATA 2025 PDF read 후 결정. 시나리오 1 (반영 됨): 본 paper 가 ATA 2025 framework 보강. 2 (부분 반영): orthogonal molecular axis 강조. 3 (미반영): "ATA 2025 의 미반영 영역" 명시.

Q5

"Cell Reports Medicine ★ vs JCI Insight + Nat Commun dual reach — 5/5 결정 필요?"

Cell Rep Med 1순위 (mechanism + translational dual fit + Cell Press Brief Report 형식).
JCI Insight reach 는 5-pillar evidence chain 정당화. Nat Commun reach 는
unsupervised discovery framing 강조 시 가능.

Colophon

Set in Cormorant Garamond +
Newsreader + JetBrains Mono.
Burgundy / Gold / Sage
palette. Single HTML · CSS
inline.

Source & provenance

Source ·
project/manuscript_v8/ ·
17 files (v1–v3 drafts).
Memory · v17_marathon_mode +
v17_8gene_audit +
v17_land2016_cite_save.

Scope

Paper 1 · cancer dark matter
(BRAF/RAS-negative PTC).
Paper 2 / 3 territory terms
· 0 plain-text occurrences.